Dry eye disease treatment: a systematic review of published trials and a critical appraisal of therapeutic strategies.

Dry eye disease (DED) treatment is an area of increasing complexity, with the emergence of several new treatment agents in recent years. Evaluation of the efficacy of these agents is limited by heterogeneity in outcomes definition and the small number of comparative studies. We provide a systematic review of clinical trials (CTs) related to DED treatment and a critical appraisal of CT public databases. CT reports obtained from eight databases were reviewed, as well as public free-access electronic databases for CT registration. Data evaluation was based on endpoints such as symptoms, Schirmer test, ocular surface staining scores, recruitment of patients, type and efficacy of the drug, and the design and site of performance of the study. Forty-nine CTs were evaluated involving 5,189 patients receiving DED treatment. Heterogeneity in study design prevented meta-analysis from yielding meaningful results, and a descriptive analysis of these studies was conducted. The most frequent categories of drugs for DED in these studies were artificial tears, followed by anti-inflammatory drugs and secretagogues. Although 116 studies have been completed, according to the registration database for clinical trials, only 17 of them (15.5%) were published. Out of 185 registered CTs related to DED, 72% were performed in the USA. The pharmaceutical industry sponsored 78% of them. The identification of effective DED treatment strategies is hindered by the lack of an accepted set of definitive criteria for evaluating disease severity.